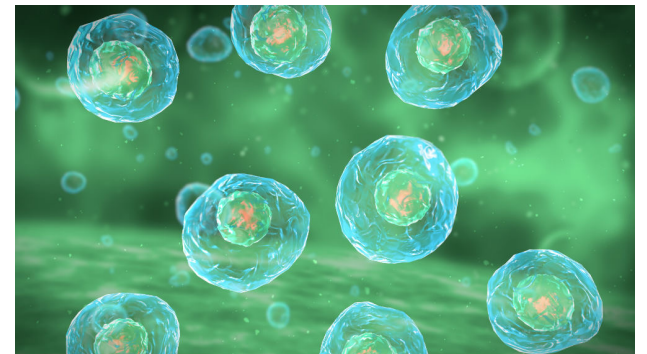


Harnessing the power of single-cell large language models with parameter-efficient fine-tuning using scPEFT



Single cell Large Language Models (scLLMs)

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scGPT: toward building a foundation model for single cell multi-omics using generative AI

[Haotian Cui](#), [Chloe Wang](#), [Hassaan Maan](#), [Kuan Pang](#), [Fengning Luo](#), [Nan Duan](#) & [Bo Wang](#) 

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A cell atlas foundation model for scalable search of similar human cells

[Graham Heimberg](#) , [Tony Kuo](#), [Daryle J. DePianto](#), [Omar Salem](#), [Tobias Heigl](#), [Nathaniel Diamant](#), [Gabriele Scalia](#), [Tommaso Biancalani](#), [Shannon J. Turley](#), [Jason R. Rock](#), [Héctor Corrada Bravo](#), [Josh Kaminker](#) , [Jason A. Vander Heiden](#)  & [Aviv Regev](#) 

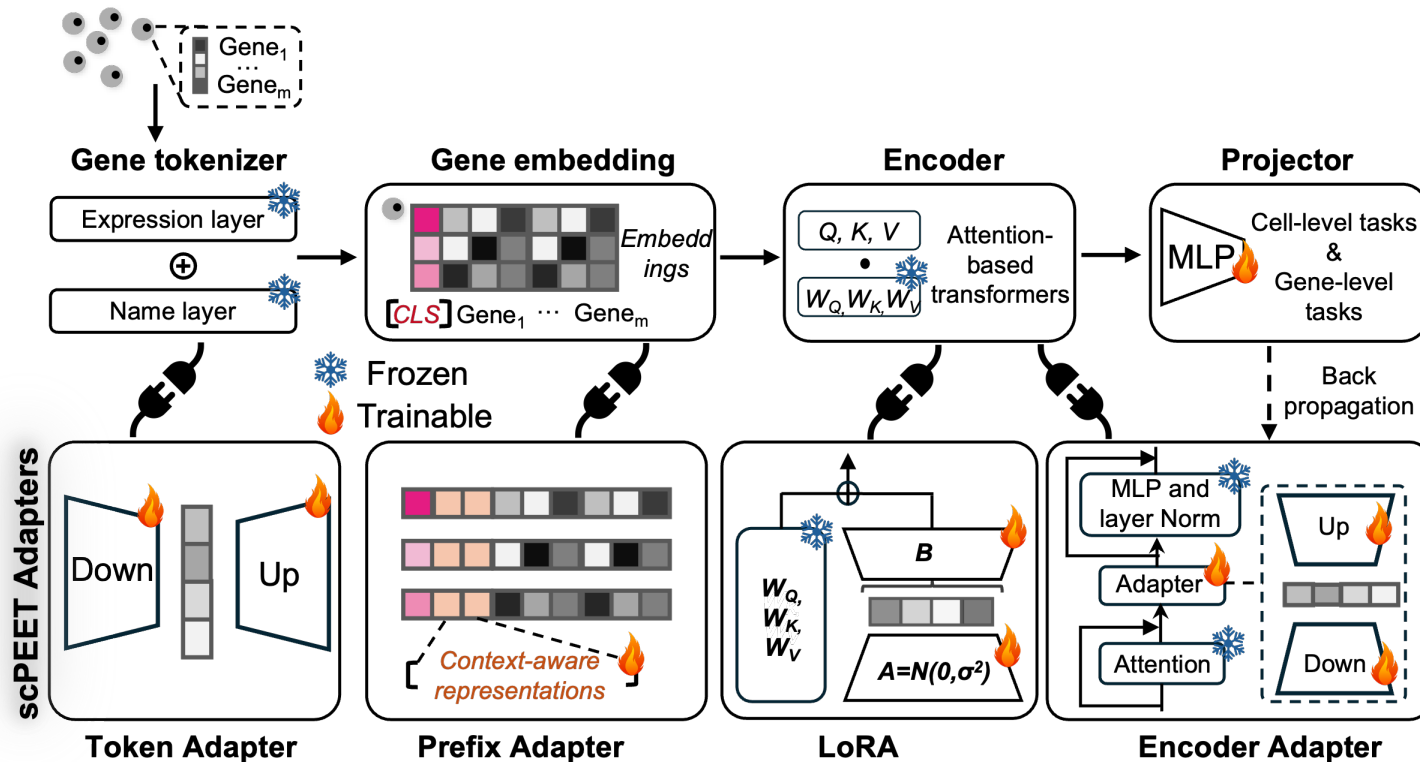
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Large models for single-cell RNA-seq data

- Trained on millions of cells
- Can be applied to various downstream tasks
- May be better in signal-to-noise ratios, batch effects...
- Do not perform well on zero-shot (Kedzierska et al., bioRxiv 2023.10.16.561085)
- Do not have emerging intelligence (foundation models?)
- Need adaptation and prompts to perform well

Parameter-Efficient Fine-Tuning scLLMs (scPEFT)

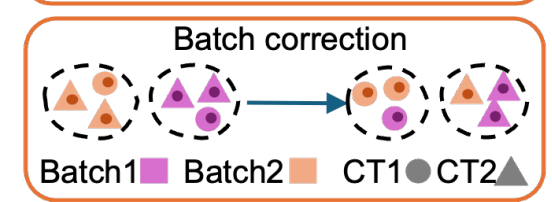
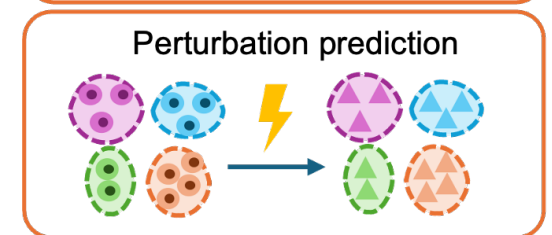
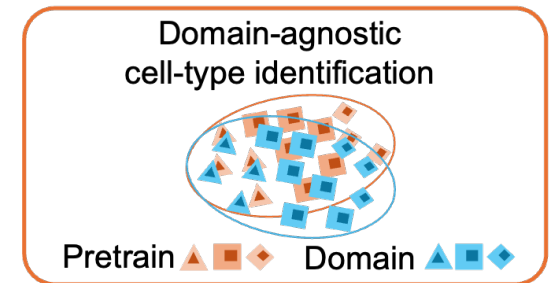
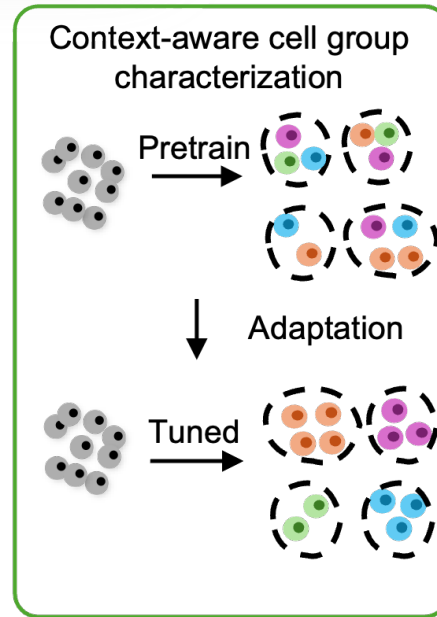
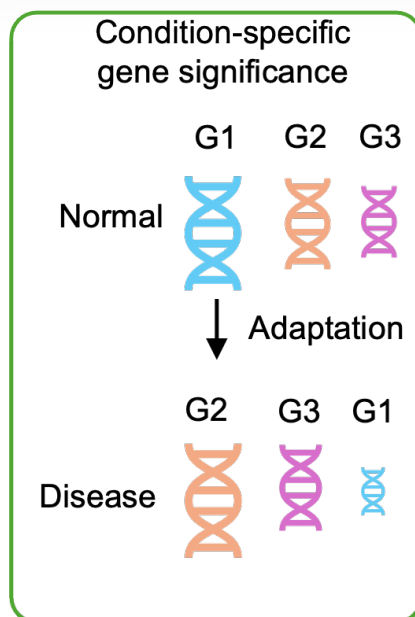
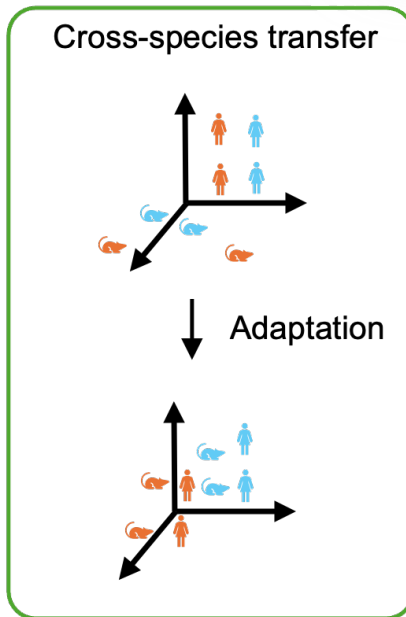


✓ Prevent catastrophic forgetting

✓ Minimize computational demands

Enhance scLLMs in out-of-context applications

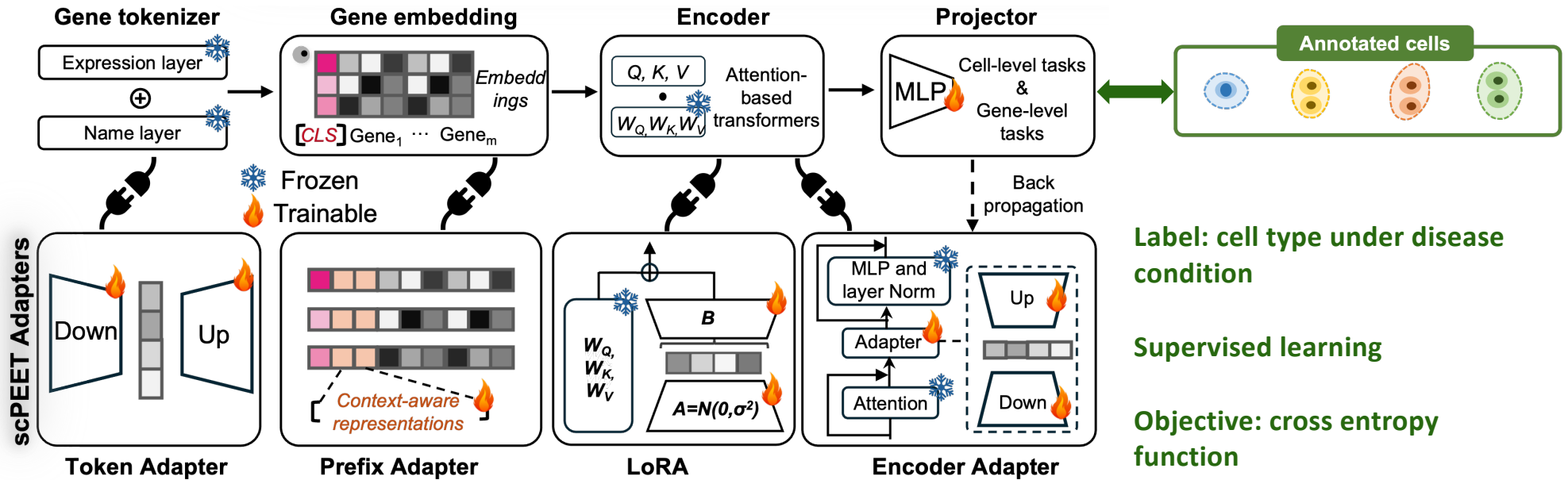
scPEET downstream applications



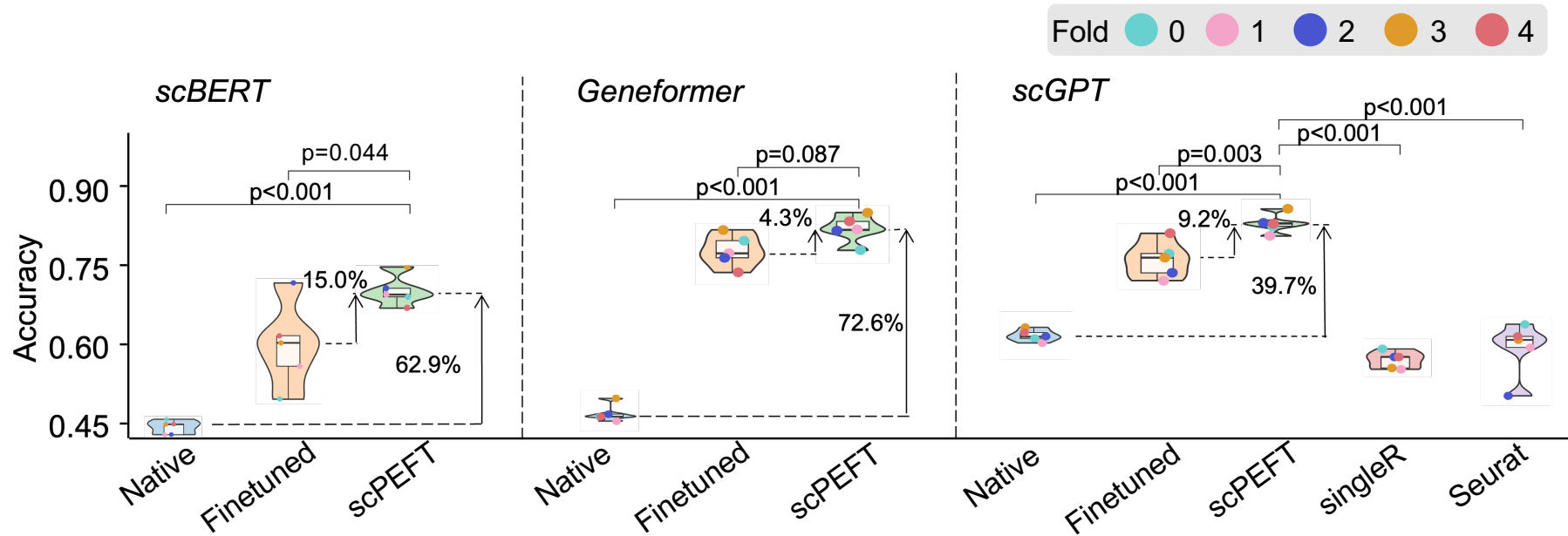
Enable

Enhance

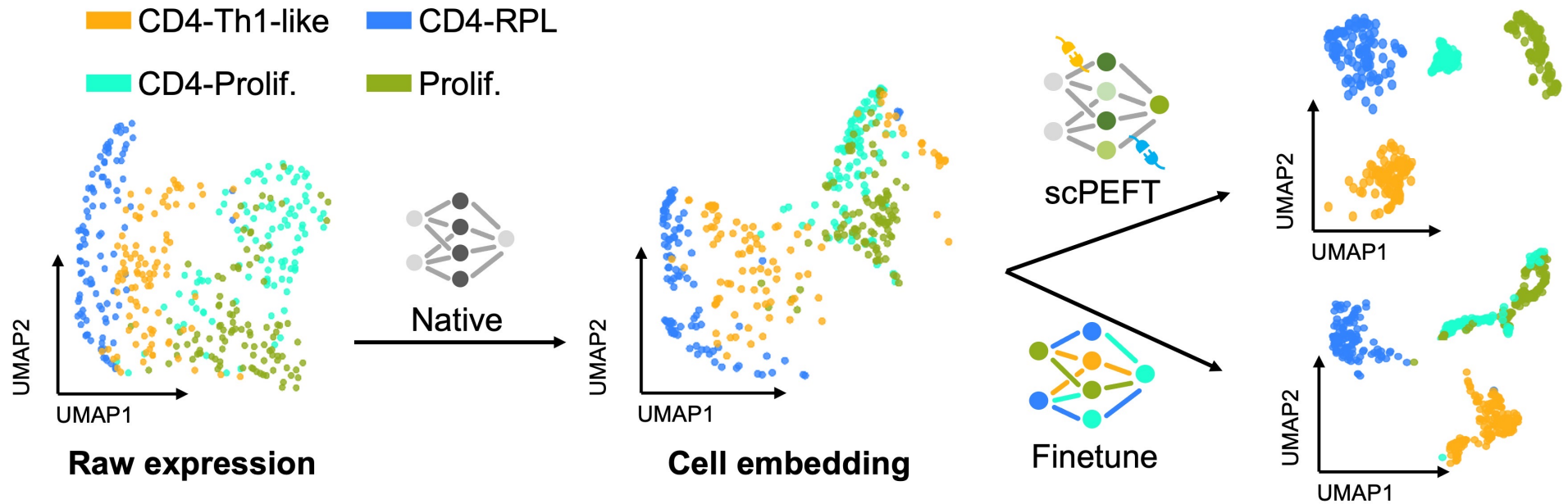
Workflow of cell type identification



Benchmarking on NSCLC dataset

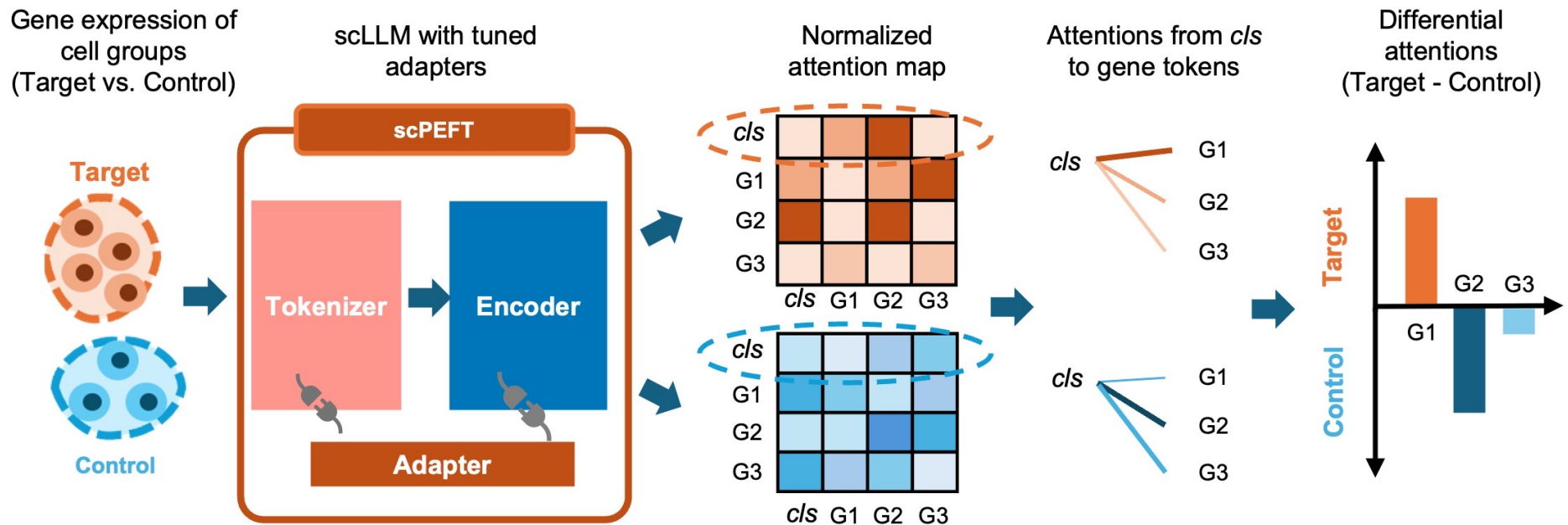


UMAP illustration of catastrophic forgetting

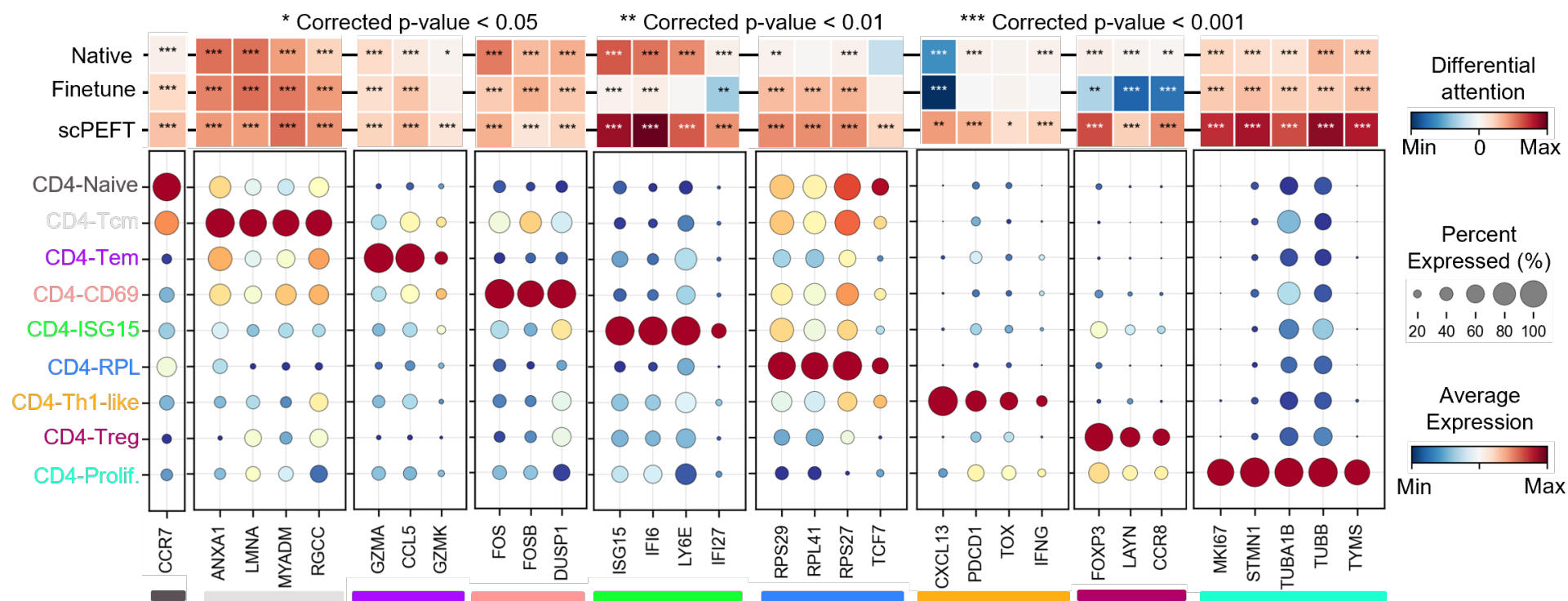


- Randomly extract 10% testing cells from these types

Differential attention analysis of scPEFT



Validation of scPEFT's differential attention scores



Markers source from: Liu, B., Hu, X., Feng, K. et al. Temporal single-cell tracing reveals clonal revival and expansion of precursor exhausted T cells during anti-PD-1 therapy in lung cancer. *Nat Cancer* 3, 2022:108–12.

Condition-specific marker discovery

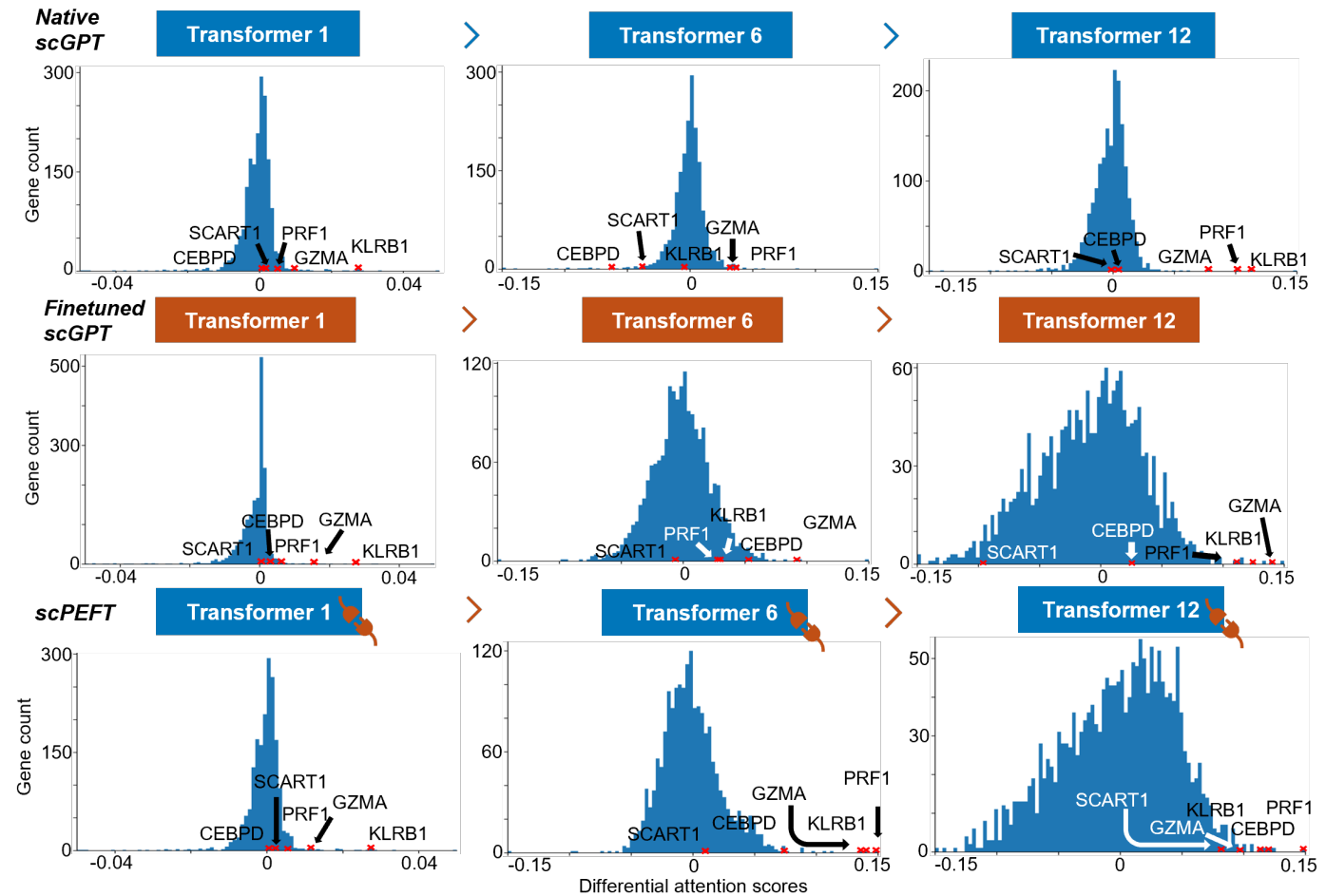
Dataset: COVID

Cell states:

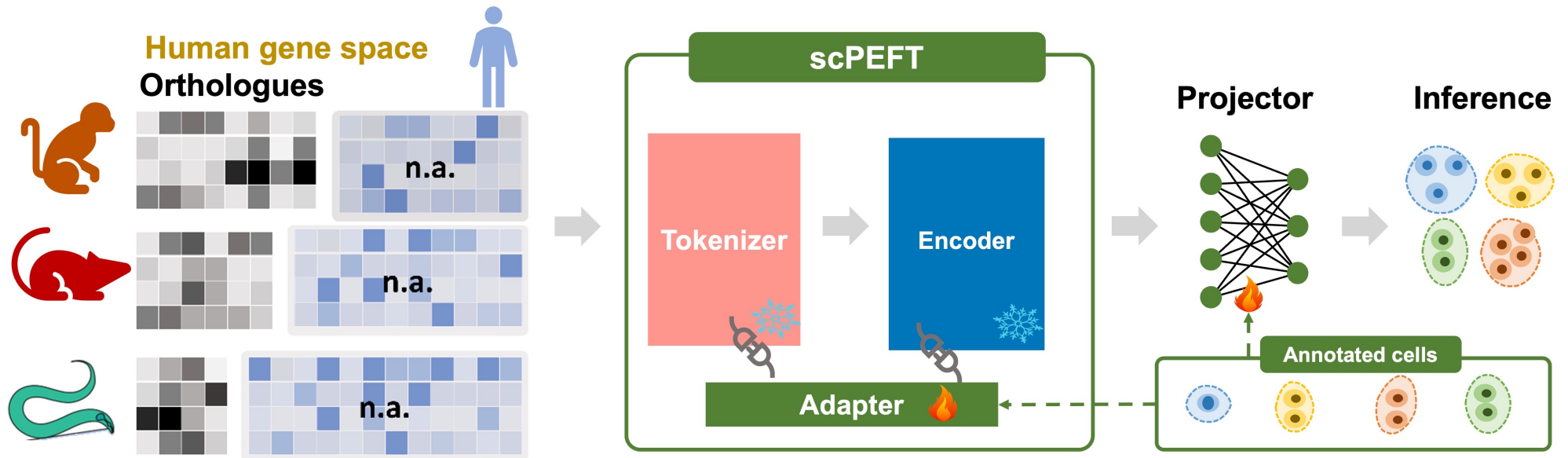
Effector Memory CD8+ T vs
Memory CD8+ T cells

Probes:

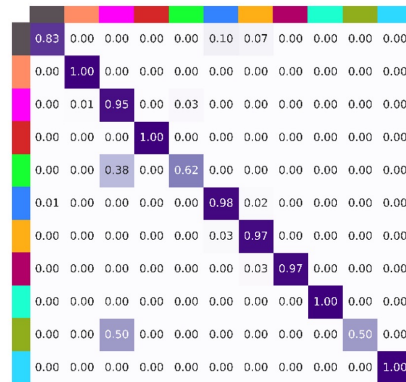
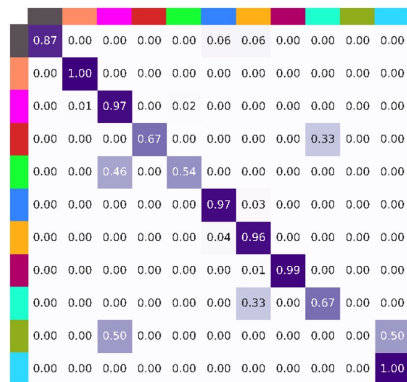
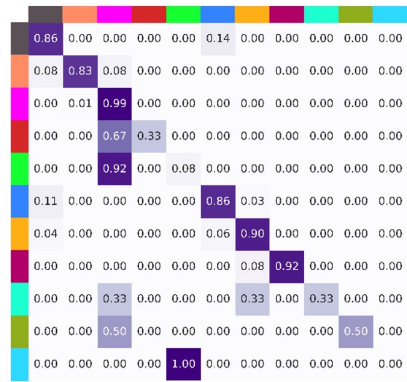
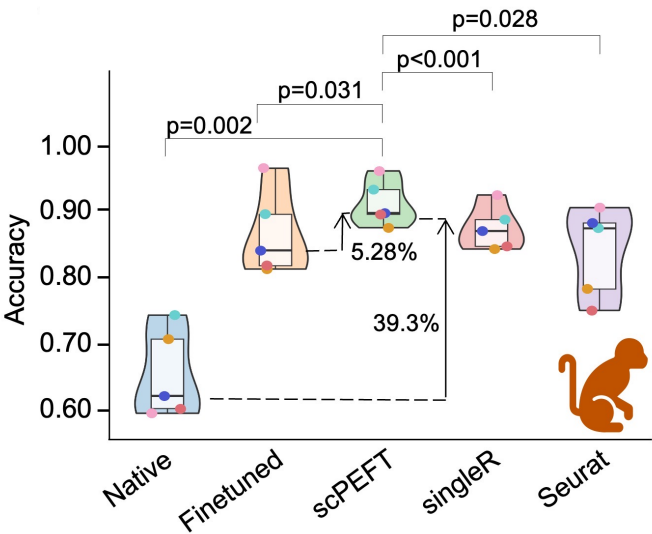
- *KLRB1*, *GZMA*, *PRF1*: Key effector molecules
- *CEBPD* and *SCART1*: Effector-function-associated genes



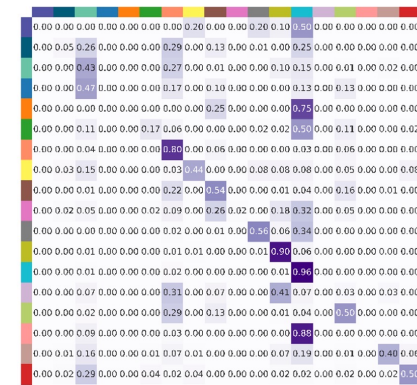
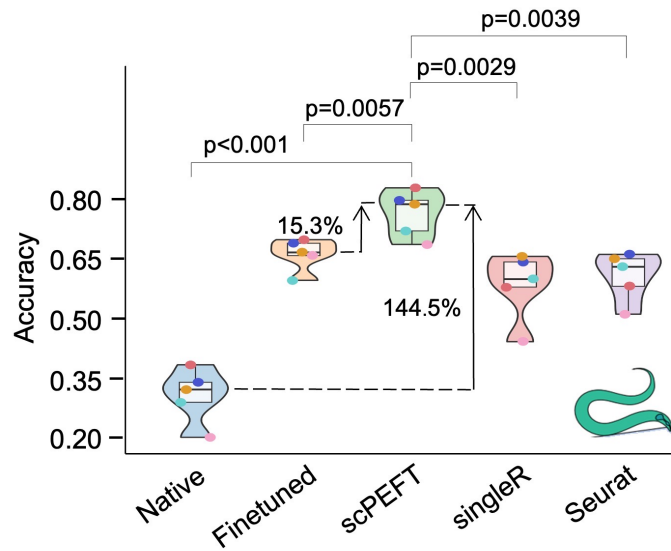
Workflow of cross-species cell type identification



Adaptation of scLLMs to macaque cell identification

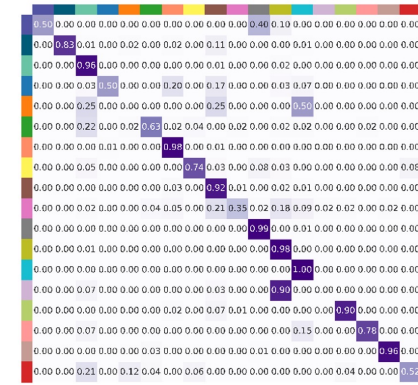


Adaptation of scLLMs to *C. elegans* cell identification



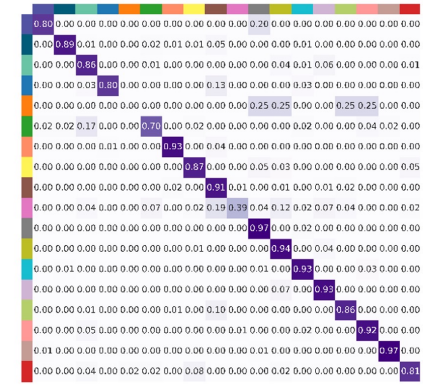
Native

coelomocyte endocrine expithelium gonad seam
hypodermis intestine various
valve



Finetuned

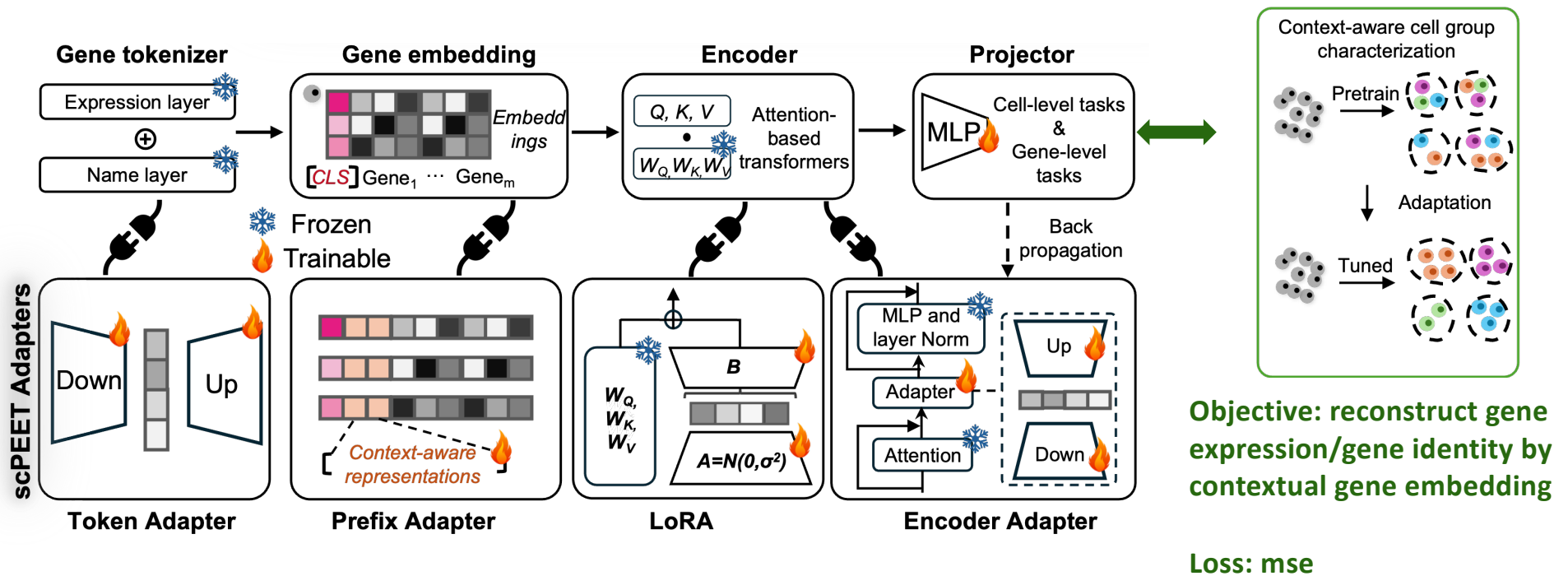
excretory mesoderm vulva



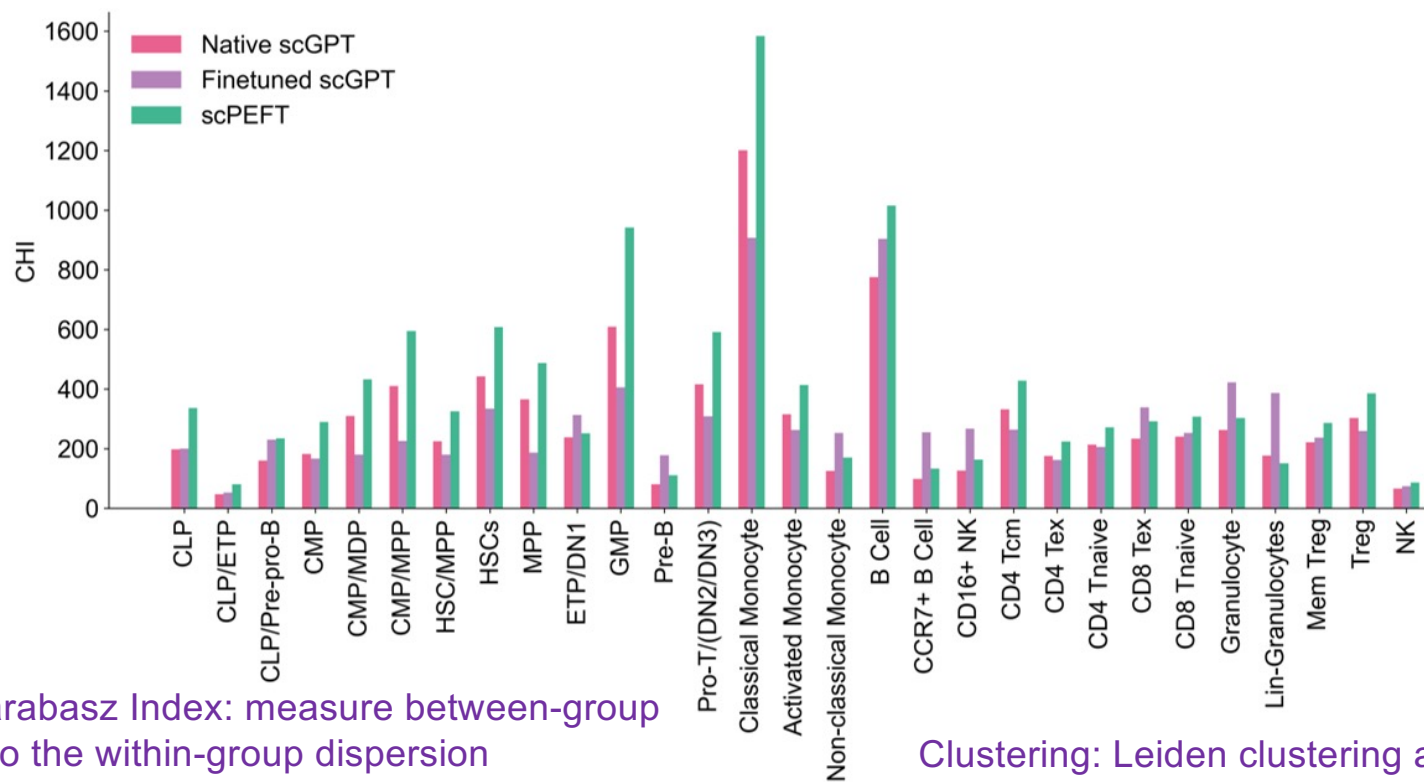
scPEFT

germline muscle gland neuron pharynx
glia

Adaption of scPEFT in unsupervised manner



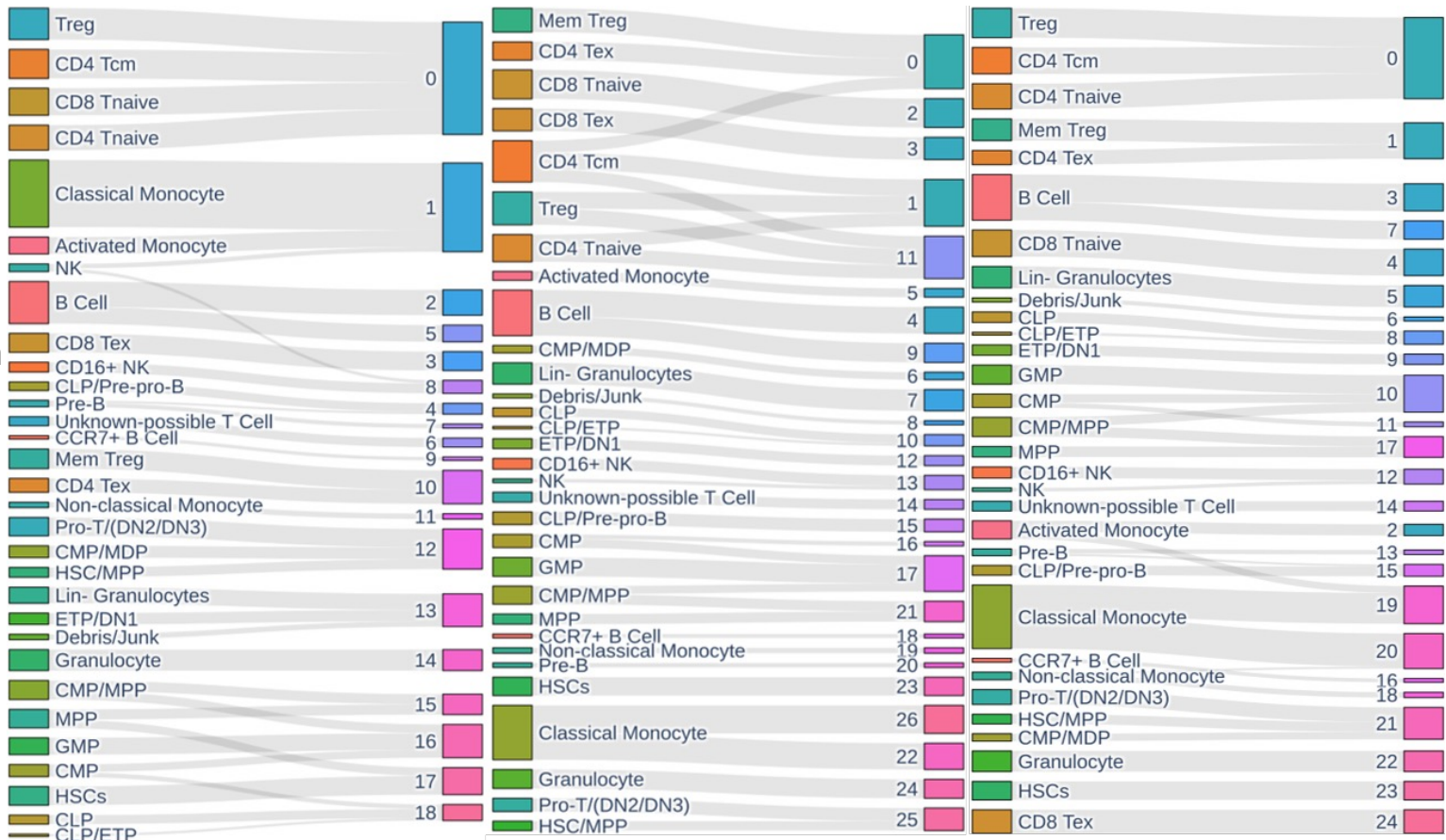
Cell clustering from adapted models



Calinski-Harabasz Index: measure between-group separation to the within-group dispersion

Clustering: Leiden clustering at a resolution of 1.5

Cell groups annotated by surface protein expression

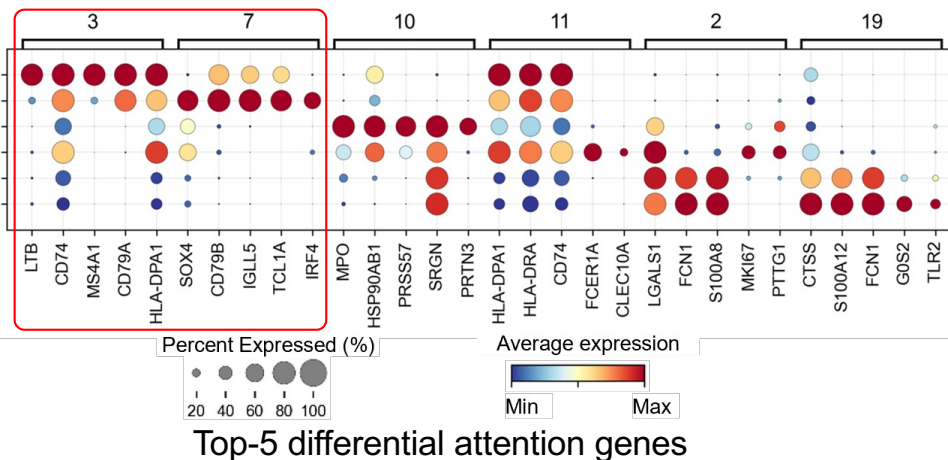
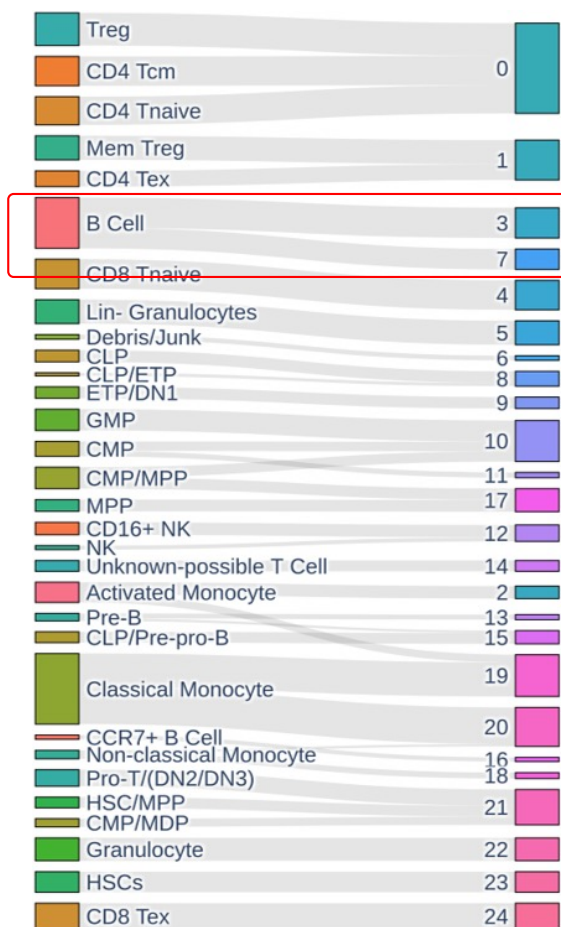


Cell groups clustered by cell embedding

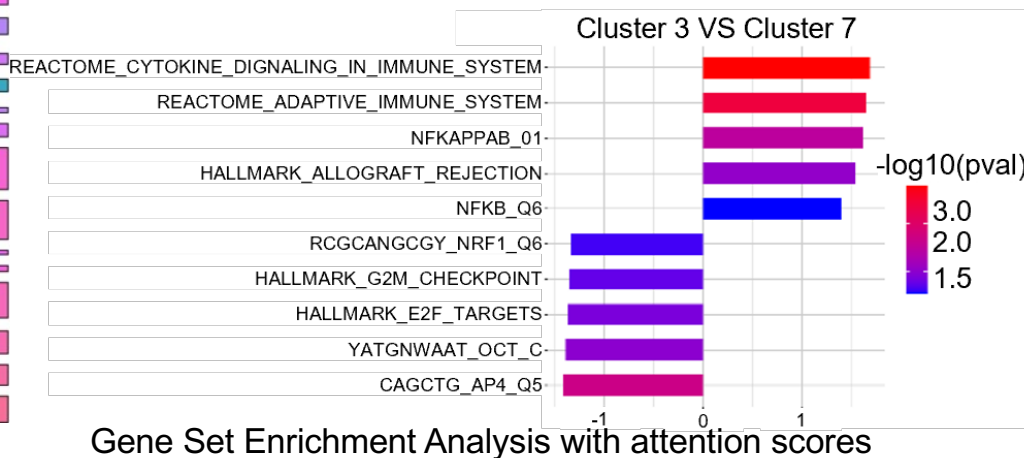
Native scGPT

Finetuned scGPT

scPEFT



Cluster 3 enriched for *NFkB* target genes and HLA molecules: a recently **developed B cells**



Cluster 7 showed enrichment for pro-growth and developmental pathways (*E2F* and *AP4* targets, *SOX4*, *IRF4*): **activated or memory B cells**